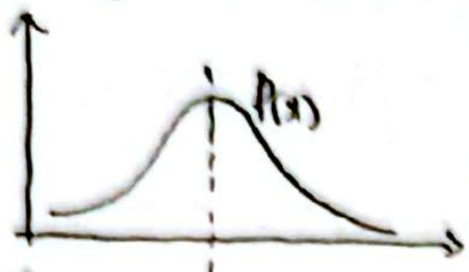
Lecture 4 : Mixture Models & Clustering :

- Why mixture/clustering?



单峰：

- Only a single mode in current probability density functions. (1个中心，1种尺度).

o What about more complicated data? (现实数据往往是多峰、多种群)

one Gaussian doesn't fit well. (e.g. ~~Bring~~ Bridge sensor to measure weight of a passing (scooters / cars / trucks) vehicle.

o GMM 模型 (Gaussian Mixture Model) :

1. Two random variable (r.v) : Z, X.

① Z = (hidden state) $Z \in \{1, 2 .. k\}$ prior probability $\boxed{P(Z=j) = \pi_j}$, $\sum_j \pi_j = 1$.

Prior :

② X = (observation) : conditional on $Z=j$

- $p(x|z=j) = N(x|\mu_j, \sigma_j^2)$ [或 $N(x|\mu_j, \Sigma_j)$].

2. Generate process (Generative process).

- Sample $z \sim p(z)$ We only see X not Z. (incomplete data).
- Sample $x|z \sim p(x|z)$

3. 边缘分布 (Mixture distribution) :

$$p(x) = \sum_z p(x|z)p(z) = \sum_j p(x|z=j)p(z=j)$$

$$p(x) = \sum_{j=1}^{k} \underbrace{\pi_j}_{\text{component weight}} \underbrace{N(x|\mu_j, \sigma_j^2)}_{\text{mixture component}}$$ (weighted sum of Gaussian pdfs).

o 用GMM做聚类 (Clustering with GMM) :

Given $D = \{x_1 .. x_N\}$ assume k clusters, each cluster is a Gaussian dist.

Estimate a GMM :

- cluster center μ_j & extent σ_j^2
- prior probability π_j of each cluster
- assignment of sample x_i to cluster z_i (分配).

Hard 分配 = 每个点只属于1个 cluster (z_i 具体)

Soft 分配 = 每个点对每个 cluster 都有1个概率.

Antoni's hack (硬分配改成软分配，假设 z 不是 hidden state，而是待优化变量).

Data $\mathcal{X} = \{x_1 .. x_N\}$ $z_i \in \{1 .. k\}$ cluster assignment of ith sample.

* Maximize joint log likelihood of (x_i, z_i) :

$$\hat{\theta} = \arg\max_{\theta, z} \sum_{i=1}^{N} \log p(x_i, z_i) = \arg\max_{\theta, z} \sum_{i=1}^{N} \left(\log p(x_i|z_i) + \log p(z_i)\right)$$

where $p(z_i) = \pi_{z_i}$

$p(x_i|z_i) = N(x_i | \mu_{z_i}, \sigma_{z_i}^2)$

- Indicator variable trick. $z_{ij} = \begin{cases} 1 & z_i = j \\ 0 & \text{otherwise} \end{cases}$. 且 $\sum_j z_{ij} = 1$ $\Rightarrow p(z_i) = \prod_{j=1}^{k} \pi_j^{z_{ij}}$

$\Rightarrow p(x_i|z_i) = \prod_{j=1}^{k} N(x_i|\mu_j, \sigma_j^2)^{z_{ij}}$ (代入)

$$\boxed{\hat{\theta} = \arg\max_{\theta, z} \sum_i \left(\log \prod_j N(x_i|\mu_j, \sigma_j^2)^{z_{ij}} + \log \prod_{j=1}^{k} \pi_j^{z_{ij}}\right) = \arg\max_{\theta, z} \sum_{i=1}^{N} \sum_{j=1}^{k} z_{ij} \left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)}$$

Expectation Maximization (EM) Algorithm :

Maximum Likelihood Estimate when there are hidden variables :

X = observation r.v. $\quad p(x,z) = p(x|z)p(z) \Rightarrow p(x) = \sum_z p(x|z)p(z)$.

Z = hidden r.v.

$X = \{x_1 - x_N\}$ ← see. $Z = \{z_1 .. z_N\}$ don't see. $\quad D = \{X, Z\}$.

final $\theta = \arg\max_\theta \log p(X) = \arg\max_\theta \log \underbrace{\sum_z p(x|z)p(z)}_{\text{sum inside log.}}$

- Given $\{x, z\}$. the problem is easy = $\underbrace{\hat{\theta} = \arg\max \log p(x, z)}$ (Antoni hack ☺).
- Guess the values = (probabilistically) =

  1) select expect value of $z \Rightarrow \hat{z}$.
  2) maximize $\log p(x, \hat{z})$ to get new model $\hat{\theta}$
  3) repeat.

- Forming =

  0) select a initial $\hat{\theta}(\text{old})$
  1) E-step $Q(\theta, \hat{\theta}^{(\text{old})}) = E_{z|x\,\theta^{\text{old}}}[\log p(x, z|\theta)]$
  2) M-step $\hat{\theta}^{(\text{new})} = \arg\max_\theta Q(\theta, \hat{\theta}^{(\text{old})})$
  3) $\hat{\theta}^{\text{old}} \leftarrow \theta^{\text{new}}$ repeat (1) & (2)

EM for GMS :

Complete data Likelihood :

$$\log p(x, z) = \sum_{i=1}^{N}\sum_{j=1}^{K} z_{ij}\left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)$$

- E-step (responsibility) = $Q(\theta, \hat{\theta}^{(\text{old})}) = E_{z|x, \hat{\theta}^{(\text{old})}}[\log p(x, z|\theta)]$

$$= \sum_{i=1}^{N}\sum_{j=1}^{K} \underbrace{E_{z|x,\hat{\theta}^{\text{old}}}[z_{ij}]}_{\hat{z}_{ij}} \left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)$$

$\hat{z}_{ij} = E_{z|x\,\theta^{\text{old}}}[z_{ij}]$

$$= \frac{\hat{\pi}_j N(x_i|\hat{\mu}_j, \hat{\sigma}_j)}{\sum_{l=1}^{K} \hat{\pi}_l N(x_i|\hat{\mu}_l, \sigma_l^2)}$$

$= p(z_i = j | x_i, \hat{\theta}^{\text{old}})$

- M-step = ☺ = $z_{ij} \to \hat{z}_{ij}$

$\hat{N}_j = \sum_{i=1}^{N} \hat{z}_{ij}$

$\hat{\mu}_j = \frac{1}{\hat{N}_j}\sum_{i=1}^{N} \hat{z}_{ij} x_i$

$\hat{\sigma}_j^2 = \frac{1}{\hat{N}_j}\sum_{i=1}^{N} \hat{z}_{ij}(x_i - \hat{\mu}_j)^2$

$\hat{\pi}_j = \hat{N}_j / N$

#: multiple solution =

Lecture 1 Non parameter Density Estimation:

交替最大化(二步循环):(1) 固定θ、更新z 选择最优簇(硬分配).

- Given $\theta = \{\mu_j, \sigma_j^2, \pi_j\}_{j=1}^{K=1}$ fixed, find z_i that maximizes.

(2) 固定z更新θ、求解参数 μ, Σ, π: Given $\{z_i\}_{i=1}^{N}$ fixed, find θ that maximizes.

(repeat (1) (2) until converge).

$$\boxed{\hat{\theta} = \arg\max_{\theta, z} \sum_{i=1}^{N}\sum_{j=1}^{N} z_{ij}\left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)}$$

(1) Find z_i's $= \hat{z} = \arg\max_{z} \sum_i \sum_j z_{ij}\left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)$

$$\Rightarrow z_i = \arg\max_{z_i} \sum_j z_{ij}\left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)$$

$\{0,1\}$

$$\Rightarrow \boxed{z_i = \arg\max_{j \in \{1,\dots,k\}} \log(N(x_i|\mu_j, \sigma_j^2) + \log \pi_j}$$

(2) Find θ = · Mean = $\{\hat{\mu}_j\}_{j=1}^{k} = \arg\max_{\{\mu_j\}} \sum_i \sum_j z_{ij}\left(\log N(x_i|\mu_j, \sigma_j^2) + \log \pi_j\right)$

$$\Downarrow$$

$$\mu_j = \arg\max_{\mu_j} \sum_i \left(\log \underline{N(x_i|\mu_j, \sigma_j^2)} + \log \pi_j\right)$$

$$\Rightarrow \mu_j = \sum_i z_{ij}(x_i - \mu_j)^2$$

argmax

$$\Rightarrow \frac{\partial}{\partial \mu_j} = -\sum_i z_{ij}\, 2(x_i - \mu_j)(-1) = 0$$

$$\rightarrow \sum_i z_{ij} x_i - \sum z_{ij}\mu_j = 0$$

$$\Rightarrow \boxed{\hat{\mu}_j = \frac{1}{\sum_{i=1}^{N} z_{ij}} \cdot \sum_{i=1}^{N} z_{ij} x_i}$$

sum of samples assigned to cluster j

#: Samples assigned to cluster j

similarly = $\boxed{\hat{\sigma}_j^2 = \frac{\sum_{i=1}^{N} z_{ij}(x_i - \mu_j)^2}{\sum_{i=1}^{N} z_{ij}}}$

Variance of samples assigned to j.

$$\boxed{\hat{\pi}_j = \frac{\sum_{i=1}^{N} z_{ij}}{N}}$$

fraction of samples assigned to cluster j.

The 2 steps converges since each step maximizes.

- if $\sigma_j^2 = 1$ $\pi_j = \frac{1}{k}$ ⇒ k-means Algorithm (Lloyd's algorithm)
- Not maximize data likelihood $\log p(x)$!
- Need to initialize θ or Z somehow.